



MY SQL PROJECT

DATA SETS USED: TATA_TB1, TATA_TB2



1. Write a query(s) to calculate the total count of rows in both tables?

SQL

```
SELECT COUNT(*) FROM TATA_TB1;  
SELECT COUNT(*) FROM TATA_TB2;
```

2. Write a query to calculate the total count of customers?

SQL

```
SELECT COUNT(DISTINCT CUSTOMERNAME) AS TTLCUSTOMERS FROM TATA_TB1;
```

3. Write a query to fetch the minimum order date and maximum order date?

SQL

```
SELECT MIN(ORDERDATE) AS OLDORDERDATE, MAX(ORDERDATE) AS LATESTORDERDATE FROM TATA_TB1;
```

4. Write query to get years in table?

SQL

```
SELECT DISTINCT(YEAR(ORDERDATE)) AS YRS FROM TATA_TB1;
```

5. Write a query to get the no. of regions and display the region names?

SQL

```
SELECT COUNT(DISTINCT REGION) REGON FROM TATA_TB1;  
SELECT DISTINCT REGION FROM TATA_TB1;
```

6. Write a query to get the no. of countries and display the country names?

SQL

```
SELECT COUNT(DISTINCT COUNTRY) CNTRY FROM TATA_TB1;  
SELECT DISTINCT COUNTRY FROM TATA_TB1;
```

7. Write a query to get the no. of states and display the state names?

SQL

```
SELECT COUNT(DISTINCT STATE) AS STAT FROM TATA_TB1;  
SELECT DISTINCT STATE FROM TATA_TB1;
```

8. Write a query to get the no. of cities and display the city names?

SQL

```
SELECT COUNT(DISTINCT CITY) AS CTY FROM TATA_TB1;  
SELECT DISTINCT CITY FROM TATA_TB1;
```

9. Write a query to calculate the total count of customers?

SQL

```
SELECT COUNT(DISTINCT CUSTOMERNAME) CSTMRNAME FROM TATA_TB1;
```

10. Write a query to calculate the total count of products?

SQL

```
SELECT COUNT(DISTINCT PRODUCTNAME) AS PRDCTNAME FROM TATA_TB2;
```

11. Write a query to calculate total sales, total profit and total order quantity?

SQL

```
SELECT SUM(SALES) AS TTLSALES, SUM(PROFIT) AS TTLLPROFIT, SUM(ORDERQUANTITY) AS TTLORDERQTY  
FROM TATA_TB2;
```

12. Write a query to calculate the total sales amount for each category.

Display the category, total sales, and total order qty and order by total sales from highest to lowest?

SQL

```
SELECT CATEGORY, SUM(SALES) AS TTLSALES, SUM(ORDERQUANTITY) AS TTLORDERQTY FROM TATA_TB2  
GROUP BY CATEGORY ORDER BY TTLSALES DESC;
```

13. Write a query to calculate the total profit amount for each category. Display the category, total profit, and total order qty and order by total profit from highest to lowest?

SQL

```
SELECT CATEGORY, SUM(PROFIT) AS TTLPROFIT, SUM(ORDERQUANTITY) AS TTLORDERQTY FROM TATA_TB2  
GROUP BY CATEGORY ORDER BY TTLPROFIT DESC;
```

14. Write a query to fetch the subcategories where total sales are greater than 100000?

SQL

```
SELECT SUBCATEGORY, SUM(SALES) AS TTLSALES FROM TATA_TB2 GROUP BY SUBCATEGORY HAVING  
TTLSALES > 100000 ORDER BY TTLSALES DESC;
```

15. Write a query to fetch the products where total profit is greater than 1000?

SQL

```
SELECT PRODCUTNAME, SUM(PROFIT) AS TTLPROFIT FROM TATA_TB2 GROUP BY PRODCUTNAME HAVING  
TTLPROFIT > 1000 ORDER BY TTLPROFIT DESC;
```

16. Write a query to get the total sales and total profit for Office Supplies category?

SQL

```
SELECT SUM(SALES) AS TTLSALES, SUM(PROFIT) AS TTLPROFIT FROM TATA_TB2  
WHERE CATEGORY = "OFFICE SUPPLIES";
```

17. Write a query to get the total sales and total profit for Furniture category and the Tables and Bookcases sub-categories ?

SQL

```
SELECT SUM(SALES) AS TTLSALES, SUM(PROFIT) AS TTLPROFIT FROM TATA_TB2 WHERE CATEGORY = "FURNITURE"  
AND SUBCATEGORY IN ("TABLES","BOOKCASES");
```

18. Write a query to get the total sales and total profit for Technology category and the Accessories, Copiers, Phones sub-categories ?

SQL

```
SELECT SUM(SALES) AS TTLSALES, SUM(PROFIT) AS TTLPROFIT FROM TATA_TB2 WHERE CATEGORY = "TECHNOLOGY" AND  
SUBCATEGORY IN ("ACCESSORIES", "COPIERS", "PHONES");
```

19. Write a query to get total sales and total profit by Region, Segment?

SQL

```
SELECT REGION,SEGMENT,SUM(SALES) AS TTLSALES,SUM(PROFIT) AS TTLPROFIT FROM TATA_TB1 A JOIN  
TATA_TB2 B ON A.ORDERID = B.ORDERID GROUP BY REGION,SEGMENT ORDER BY REGION;
```

20. Write a query to get total sales and total profit by Country, State and city?

SQL

```
SELECT COUNTRY,STATE,CITY,SUM(SALES) AS TTLSALES,SUM(PROFIT) AS TTLPROFIT FROM TATA_TB1 A JOIN  
TATA_TB2 B ON A.ORDERID = B.ORDERID GROUP BY COUNTRY,STATE,CITY ORDER BY COUNTRY;
```

21. Write a query to get total sales and total orderqty by CustomerName?

SQL

```
SELECT CUSTOMERNAME, SUM(SALES) AS TTLSALES, SUM(ORDERQUANTITY) AS TTLORDERQTY FROM TATA_TB1 A JOIN  
TATA_TB2 B ON A.ORDERID = B.ORDERID GROUP BY CUSTOMERNAME ORDER BY TTLSALES DESC;
```

22. Identify the top 5 products with the highest sales (by sales amount).
Show the product name, total sales, and total qty?

SQL

```
WITH CTE AS(SELECT PRODUCTNAME, SUM(SALES) AS TTLSALES, SUM(ORDERQUANTITY) AS TTLORDERQTY, DENSE_RANK()  
OVER(ORDER BY SUM(SALES) DESC) AS RNK FROM TATA_TB1 A JOIN TATA_TB2 B ON A.ORDERID = B.ORDERID GROUP BY  
PRODUCTNAME) SELECT * FROM CTE WHERE RNK <=5;
```

23. Write a query to get total sales by City having sales greater than 15000?

SQL

```
SELECT CITY, SUM(SALES) AS TTLSALES FROM TATA_TB1 A JOIN TATA_TB2 B ON A.ORDERID = B.ORDERID  
GROUP BY CITY HAVING TTLSALES > 15000 ORDER BY TTLSALES DESC;
```

24. Write a query to get total sales by CustomerName having sales greater than 3000?

SQL

```
SELECT CUSTOMERNAME, SUM(SALES) AS TTLSALES FROM TATA_TB1 A JOIN TATA_TB2 B  
ON A.ORDERID = B.ORDERID  
GROUP BY CUSTOMERNAME HAVING TTLSALES > 3000 ORDER BY TTLSALES DESC;
```

25. Write a query to get total sales and total profit by shipmode?

SQL

```
SELECT SHIPMODE,SUM(SALES) AS TTLSALES, SUM(PROFIT) AS TTLPROFIT FROM TATA_TB1 A JOIN TATA_TB2 B ON  
A.ORDERID = B.ORDERID GROUP BY SHIPMODE ORDER BY TTLSALES DESC;
```

26. Write a query to get total sales for North and central region?

SQL

```
SELECT SUM(SALES) AS TTLSALES FROM TATA_TB1 A JOIN TATA_TB2 B ON A.ORDERID = B.ORDERID  
WHERE REGION IN ("NORTH", "CENTRAL");
```

27. Write a query to get total sales and total profit for Italy and Spain countries?

SQL

```
SELECT SUM(SALES) AS TTLSALES, SUM(PROFIT) AS TTLPROFIT  
FROM TATA_TB1 A JOIN TATA_TB2 B ON A.ORDERID = B.ORDERID WHERE COUNTRY IN ("ITALY","SPAIN");
```

28. Write a query to get the total sales and total profit for each year?

SQL

```
SELECT YEAR(ORDERDATE) AS YER, SUM(SALES) AS TTLSALES , SUM(PROFIT) AS TTLPROFIT  
FROM TATA_TB1 A JOIN TATA_TB2 B ON A.ORDERID = B.ORDERID GROUP BY YER ORDER BY TTLSALES DESC;
```

**29. Find the top 10 customers who spent the most across all transactions.
Display the customer name, total amount spent, and number of orders placed?**

SQL

```
WITH CTE AS(SELECT CUSTOMERNAME,SUM(SALES) AS TTLSALES, SUM(ORDERQUANTITY) AS TTLORDERQTY,  
DENSE_RANK() OVER(ORDER BY SUM(SALES) DESC) AS RNK FROM TATA_TB1 A JOIN TATA_TB2 B ON  
A.ORDERID = B.ORDERID GROUP BY CUSTOMERNAME ORDER BY TTLSALES DESC)  
SELECT * FROM CTE LIMIT 10;
```

30. Write a query to find which products are most preferred by customers based on the total sales. Display customer name, favourite product(top 3 products per each customer) and total sales on that product?

SQL

```
WITH CTE AS (SELECT CUSTOMERNAME,PRODUCTNAME,SUM(SALES) AS TTLSALES,  
ROW_NUMBER() OVER(PARTITION BY CUSTOMERNAME ORDER BY SUM(SALES) DESC)AS RNK  
FROM TATA_TB1 A JOIN TATA_TB2 B ON A.ORDERID = B.ORDERID GROUP BY CUSTOMERNAME,PRODUCTNAME )  
SELECT * FROM CTE WHERE RNK <=3;
```

**31. Write a query to get 7th rank customer name based on total sales?
Display customer name, sales amount and rank.**

SQL

```
WITH CTE AS(SELECT CUSTOMERNAME,SUM(SALES) AS TTLSALES,DENSE_RANK() OVER(ORDER BY SUM(SALES) DESC) AS RNK  
FROM TATA_TB1 A JOIN TATA_TB2 B ON A.ORDERID = B.ORDERID GROUP BY CUSTOMERNAME)  
SELECT * FROM CTE WHERE RNK = 7;
```

**32. Write query to get total sales and total profit between year 2011 and 2013?
Display year wise total sales and total profit?**

SQL

```
WITH CTE AS (SELECT YEAR(ORDERDATE) AS YR, SUM(SALES) AS TTLSALES, SUM(PROFIT) AS TTLPROFIT FROM TATA_TB1 A JOIN  
TATA_TB2 B ON A.ORDERID = B.ORDERID GROUP BY YR)  
SELECT * FROM CTE WHERE YR BETWEEN 2011 AND 2013;
```

**33. Write a query to get total sales, total profit and total order qty ?
#display country, state, category and sub-category?**

SQL

```
SELECT COUNTRY, STATE, CATEGORY, SUBCATEGORY, SUM(SALES) AS TTLSALES, SUM(PROFIT) AS TTLFPROFIT,  
SUM(ORDERQUANTITY) AS TTLORDERQTY FROM TATA_TB1 A JOIN TATA_TB2 B  
ON A.ORDERID = B.ORDERID GROUP BY 1,2,3,4 ORDER BY 5 DESC;
```

34. Write stored procedure to get top 10 customers based on total sales
If we call top 10 customers you need to display customer and sales?

SQL

```
DELIMITER &&
CREATE PROCEDURE TOP10CUST()
BEGIN
WITH CTE AS(
SELECT CUSTOMERNAME,SUM(SALES) AS TTLSALES,DENSE_RANK() OVER(ORDER BY SUM(SALES) DESC) AS RNK
FROM TATA_TB1 A JOIN TATA_TB2 B ON A.ORDERID = B.ORDERID GROUP BY CUSTOMERNAME)
SELECT * FROM CTE WHERE RNK <= 10;
END;
&&
CALL TOP10CUST();
```

35. Create a virtual table (view) by using OrderID, OrderDate, CustomerName, Region, country, sales, profit and order qty?

SQL

```
CREATE VIEW TATACUSTVIEW AS(
SELECT A.ORDERID,A.ORDERDATE,A.CUSTOMERNAME,A.REGION,A.COUNTRY,B.SALES,B.PROFIT,B.ORDERQUANTITY
FROM TATA_TB1 A LEFT JOIN TATA_TB2 B ON A.ORDERID = B.ORDERID) ;
```

36. Create a stored procedure to get region wise sales and profit?
If we call procedure, you need to display region, sales and profit?

SQL

```
DELIMITER &&
CREATE PROCEDURE REGIONSALES (
IN REGIN VARCHAR(20)
)
BEGIN
SELECT * FROM (SELECT REGION,SUM(SALES) AS TOTALSALES, SUM(PROFIT) AS TOTALPROFIT FROM
(SELECT A.REGION,A.COUNTRY,B.SALES,B.PROFIT FROM TATA_TB1 A INNER JOIN TATA_TB2 B
ON A.ORDERID = B.ORDERID) C GROUP BY REGION) D WHERE REGION = REGIN;
END;
&&
CALL REGIONSALES("CENTRAL");
```

37. Calculate YoY% sales growth?

SQL

```
WITH CTE AS(  
  SELECT YEAR(ORDERDATE) AS YR, SUM(SALES) AS C_YRSALES,  
  LAG(SUM(SALES)) OVER (ORDER BY YEAR(ORDERDATE)) AS P_YRSALES  
  FROM TATA_TB1 A JOIN TATA_TB2 B  
  ON A.ORDERID = B.ORDERID GROUP BY YR)  
SELECT YR,C_YRSALES,P_YRSALES,((C_YRSALES - P_YRSALES)/P_YRSALES) * 100 AS YoYGrowth FROM CTE;
```



THANK YOU FOR WATCHING